A
Project Report
on
CareerMatch
submitted as partial fulfillment for the award of

BACHELOR OF TECHNOLOGY DEGREE

SESSION 2025-26

in

CSE(AI&ML)

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May, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that Project Report entitled "Project Title" which is submitted by Student name in partial fulfillment of the requirement for the award of degree B. Tech. in Department of CSE (AIML) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Dr. Payal Chhabra	**Dr. Rekha Kashyap**
Assistant Professor	HOD & Dean – CSE(AI)and CSE(AI&ML)

Date:

ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the report of the B. Tech Project undertaken during B. Tech. Final Year. We owe special debt of gratitude to supervisor name, Department of CSE(AIML), KIET, Ghaziabad, for his constant support and guidance throughout the course of our work. His sincerity, thoroughness and perseverance have been a constant source of inspiration for us. It is only his cognizant efforts that our endeavors have seen light of the day.

We also take the opportunity to acknowledge the contribution of Dr. Rekha Kashyap, HOD & Dean of the Department of Computer Science & Engineering (AI/AI&ML), KIET, Ghaziabad, for his full support and assistance during the development of the project. We also do not like to miss the opportunity to acknowledge the contribution of all the faculty members of the department for their kind assistance and cooperation during the development of our project.

We also do not like to miss the opportunity to acknowledge the contribution of all faculty members, especially faculty/industry person/any person, of the department for their kind assistance and cooperation during the development of our project. Last but not the least, we acknowledge our friends for their contribution in the completion of the project.

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ABSTRACT

The transition from academic life to professional employment is often challenging for fresh graduates due to the difficulty of identifying job opportunities that match their skills and qualifications. A significant number of current job portals have users manually searching and browsing through the job portal which could result in irrelevant job search results and inefficient use of time. This issue brings to the fore the necessity to develop smart systems that may analyze profiles of candidates automatically and suggest appropriate job openings.

The CareerMatch project offers a smart web based platform which is aimed at helping fresh graduates to find relevant job positions based on their skills, educational background, and career interests. The system enables the user to post their resumes and the resumes are examined using text processing tools to obtain meaningful information like technical skills, qualifications and keywords.

The resulting information is processed through machine learning methods including Text feature extraction: Term Frequency-Inverse Document Frequency, and classification algorithms to compare profile of the candidate to available job descriptions. According to this analysis, the system suggests job roles that are most related to the profile of the user.

The application is developed as a web platform in HTML, CSS and JavaScript as the user interface, PHP as the back-end processing and MySQL as the database management system. Other features incorporated in the system include registration of users, posting of their resumes, analysis of skills, and personalized job suggestions.

The major objective of CareerMatch is to make job search easier to fresh graduates by offering relevant and personalized job recommendations. By integrating machine learning with web technologies, the platform improves the efficiency of job matching and helps candidates identify suitable employment opportunities more effectively.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
DBMS	Database Management System
ER	Entity Relationship
HTML	HyperText Markup Language
IT	Information Technology
ML	Machine Learning
MySQL	My Structured Query Language
NLP	Natural Language Processing
PHP	Hypertext Preprocessor
TF-IDF	Term Frequency – Inverse Document Frequency
UI	User Interface
UX	User Experience

CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

The rapid technological advancements and the growing competition from recent graduates have made the job search process more challenging. New graduates can struggle to secure the right job as they lack work experience, and may be unsure about the best job roles to apply for based on their skills. Conventional job seeking approaches involve candidates searching for jobs on various portals and going through many job listings, which is tedious and inefficient.

The latest hiring systems are increasingly incorporating smart technologies to enhance candidate-job matching. Data analytics, machine learning and natural language processing (NLP) are used to automatically assess candidate and job descriptions. This enables the extraction of patterns and matches between candidate skillsets and job requirements, enabling the system to recommend jobs.

Recommendation systems have been successfully applied in other industries including e-commerce, entertainment and social media. Adapting these principles to job matching can enhance the recruitment process. By leveraging user preferences, skills, and job descriptions, recommendation algorithms can offer tailored job recommendations based on the candidate's interests and skills.

A web-based platform is built using current web technologies such as HTML, CSS, JavaScript, PHP and MySQL, and machine learning techniques such as TF-IDF for textual analysis. CareerMatch streamlines the job matching process, saving time and increasing the chances of successful candidate-employer matches

1.2 PROJECT DESCRIPTION

CareerMatch or Smart Job Recommender System for College Graduates is a web based system that intends to find suitable jobs for job applicants according to their skills and qualifications. It is designed to automate the process of job matching by providing automatic job recommendations based on resumes.

The system allows the user to register, upload their resumes and check the jobs recommended by the system. Once the user has uploaded their resume, the system starts to process the resume and extract the data such as the skills, tools, programming languages and qualifications. This information is converted into representation used by the recommendation engine.

The recommendation engine compares the information on the resume with jobs. A match between the candidate and job is determined using natural language processing. This facilitates a list of jobs that suit the candidate.

The system has various components, including the user login, resume details, job details, recommendation engine, and admin module. The admin module allows the database administrator to add, edit and remove jobs from the database.

The CareerMatch system is a full-stack web application, which has a user interface that enables user interaction and a back-end that deals with data and database interaction. The database stores user details, resumes, skills extracted from resumes, as well as job opportunities, enabling fast access and management of data.

This project will help new graduates to find appropriate jobs through a job recommendation system. The automation in resume matching and job recommendation makes the job search more efficient, and helps in finding suitable jobs based on their skills and interests.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of Job Recommendation Systems

The growth of online job platforms has revolutionised the recruitment process. In the past, job seekers had to browse through newspapers, headhunters, or the websites of companies for suitable jobs. As a result of the growth of the internet and other digital technologies, job portals are one of the most popular jobs searching platforms.

A job recommendation system is a computer-based system that helps candidates to identify potential job opportunities by matching their profile. Such systems take into account candidate information including education, technical skills, work experience and personal interests to suggest job positions that are suitable for their profile. The goal of these systems is to improve the recruitment process by making it quicker and more efficient in matching candidates to jobs.

Recommended systems are also used to enhance user experience in many job portals. These systems gather and process user information to recommend jobs. The success of such systems is determined by the ability of the system to accurately understand the job seeker and the job description.

The CareerMatch system aims to enhance the job recommendation system for job seekers with limited work experience. It will provide more precise job recommendations than the traditional job portals, which rely on search engines.

2.2 Resume Parsing Techniques

Resume parsing plays a significant role in automated hiring process. It is the process of retrieving information in resumes, which are generally not structured documents like PDF or Word documents. Employers receive a lot of resumes for a single job vacancy, and it's time-consuming and inefficient to manually process these resumes. Resume parsers make use of text mining methods to identify and extract pertinent information in resumes. They may be the name of the applicant, his/her address, phone number, education history, experience, certifications and skills. The extracted data can then be stored in a structured format for further processing by computer programs.

Resume parsing often involves using Natural Language Processing (NLP). NLP enables the system to comprehend human language by extracting keywords and phrases from the resume. The system can then identify the skills and competencies relevant to the job from the resume.

Resume parsing is a critical step in the CareerMatch system for extracting candidates' skills. The skills are then passed on to the recommender system for matching the candidate with job descriptions.

2.3 Machine Learning Techniques in Recruitment Systems

Machine learning is a game-changer for recommendation systems. Machine learning can be used in recruitment matching systems to analyse a large corpus of text in resumes and jobs. They can identify skill and job requirements.

The most common method used for text analysis is Term Frequency - Inverse Document Frequency (TF-IDF). This is used to measure the importance of terms in a document, for a collection of documents. TF-IDF can be used to extract important keywords from resumes and jobs, to help the system measure the similarity between resumes and jobs.

A more common method is to use classification techniques such as Support Vector Machines (SVM) and its variants. These can categorise resumes into various jobs based on resume features. By training the system with job descriptions, it can learn what skills are needed for a particular job.

Machine learning algorithms enable job recommendation systems to automate the job matching process and improve the quality of job recommendations. This speeds up the job search process and enhances job recommendations.

2.4 Existing Online Job Portals

There are several websites that facilitate hiring and job seekers. Sites like LinkedIn, Indeed and Naukri allow users to create accounts, upload their resumes and apply for jobs. These sites have made hiring more convenient by connecting jobseekers and employers.

Although these websites have a lot to offer, they also have their limitations. The majority of job websites provide text search engines that allow users to search for jobs. This may lead to a long list of jobs that are not suitable for users' abilities.

Also, graduates may lack experience to be paired with suitable jobs. Current systems are geared towards working professionals and provide little guidance for new graduates.

The CareerMatch project is designed to address these issues for new graduates. The system will not only be able to employ manual filtering, but also examine skills in resumes to recommend suitable roles. The other issue is that new graduates do not usually have work experience, and it is challenging to have traditional systems recommend appropriate job opportunities. The available platforms are mostly focused on experienced candidates and thus the new graduates have little advice on what to do since they are new in the job market. CareerMatch project is trying to deal with these problems, paying special attention to fresh graduates. The system can produce more precise job recommendations by analyzing the skills listed on resumes instead of just relying on the manual search filters.

2.5 Summary

There is the need of intelligent recommendation systems of jobs as demonstrated in the review. Resume parsing, natural language processing and machine learning are some of the technologies that have enhanced job candidate assessment and recommendation in the hiring process. Although the current websites also offer certain job recommendations, the recommendations are usually based on a manual search and filtering. Such search methods might not necessarily make personalized suggestions, especially to individuals who have just graduated without prior work experience.

The CareerMatch system is based on these technologies, which include a combination of resume parsing and machine learning. The idea behind this approach is to enhance job recommendations hence enhancing the job seeking process by recent graduates.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 System Overview

The system, CareerMatch, an intelligent job matching system that helps recent graduates to find job opportunities according to their skills and experience. It analyzes information on the resume of the user and compares it with job descriptions within the database in order to recommend appropriate jobs.

The system is a web-based application where users are able to create accounts, post their resumes and be given recommendations on the jobs available. It applies to job matching process web technologies and text mining to automate the process.

The suggested methodology will automatically identify all the relevant information in the resumes, process the information retrieved and compare it with the job requirements through machine learning algorithms. This methodology allows the system to automate the job search process and enhance the recommendation accuracy.

3.2 System Architecture

CareerMatch is a three tiered architecture and is referred to as the presentation layer, application layer and data link layer.

1. Presentation Layer (Frontend)

This is the user interface. This is developed with HTML, CSS and JS which enables users to register, login, upload their CV's and search for jobs.

2. Application Layer (Backend)

The application layer (backend) is a server which handles requests.

3. Data Layer (Database)

This is used to store users, resumes, skills and jobs. This is stored in MySQL database.

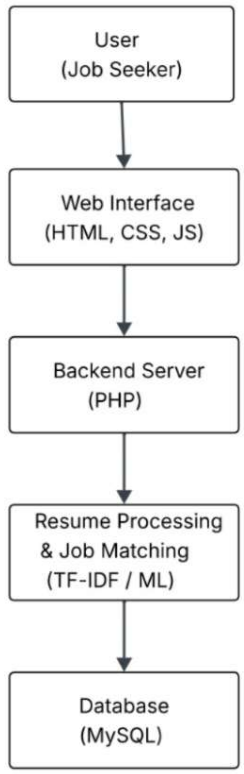

Figure 3.1 – System Architecture Diagram

Table 3.1: Tech-Stack Used in the System

Component	Technology Used	Purpose
Frontend	HTML, CSS, JS	To design the UI of the platform
Backend	PHP	To handle server-side processing
Database	MySQL	To store data and job listings
Machine Learning	Python	For resume analysis and job matching
NLP Library	NLTK	For text preprocessing and skill extraction

3.3 Resume Processing and Skill Extraction

The proposed system processes resumes. After the user has uploaded a resume, the system will read the resume and extract the text. Typically resumes contain unstructured data, and so the system will process the text of the resume looking for keywords and phrases.

The resume is first converted to a text file that the algorithm can read. Thereafter, it preprocesses the text to eliminate irrelevant characters and words. This improves the quality of the data.

Then, the system identifies the skills from the resume. This may include programming languages, tools, frameworks and other skills. These are then entered into the database and used in the job matching algorithm.

This eliminates the manual resume screening process and makes it possible for the system to process applicants' resumes.

3.4 Job Matching Using TF-IDF

The system uses text-similarity algorithms to match the best job offers for a candidate. The CareerMatch system uses one of these techniques, named TF-IDF.

TF-IDF is a statistical measure used to evaluate the relevance of a word to a document in a set of documents. In job matching, it is used to analyse the importance of keywords in resumes and job descriptions.

Here's how it works:

1. It turns resumes and job descriptions into text vectors, using the TF- IDF technique.
2. The weights of keywords are adjusted according to their importance and rarity.
3. It then measures similarity between the user skillset and job descriptions using these vectors.
4. The user is recommended jobs with the highest similarity.

3.5 System Modules

The CareerMatch system has a number of modules that cooperate to match jobs.

1. User-Authentication Module

It enables users to sign-up & log-in. This module guarantees the security and authentication of users for profile access and job matching.

2. Resume Upload Module

The user can upload a resume in a particular format (e.g., PDF, Word). The resume is analysed to extract information.

3. Resume Analysis Module

This module preprocesses and extracts skills of the resume. The information are entered into the database.

4. Job Recommendation Module

The job recommender system uses TF-IDF algorithm to compare the information on resume with job description. The scores are calculated and the user is presented with jobs.

5. Admin Module

The admin module enables the administrators to add, edit and delete jobs and descriptions. This keeps the jobs up-to-date in the database.

Table 3.2: System Modules Description

Module Name	Description
User-Registration Module	Allows users to register on the platform
Login Module	Authenticates users and allows secure access
Resume Upload Module	Enables users to upload resumes for analysis
Resume Analysis Module	Extracts skills and important information from resumes
Job Recommendation Module	Suggests relevant job roles based on extracted skills
Admin Module	Allows administrators to manage job listings

3.6 Workflow of the Proposed System

The CareerMatch process can be described as follows:

1. User registers and logs into the platform.
2. The user uploads their resume to the system.
3. The resume is scanned and keywords extracted.
4. The data is processed using text mining and machine learning algorithms.
5. The profile is matched against job descriptions.
6. The system provides a list of suitable job recommendations.

This process helps users get job recommendations based on their skills and interests.

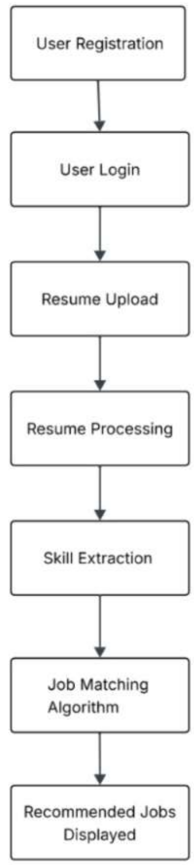

Figure 3.2 – System Workflow Diagram

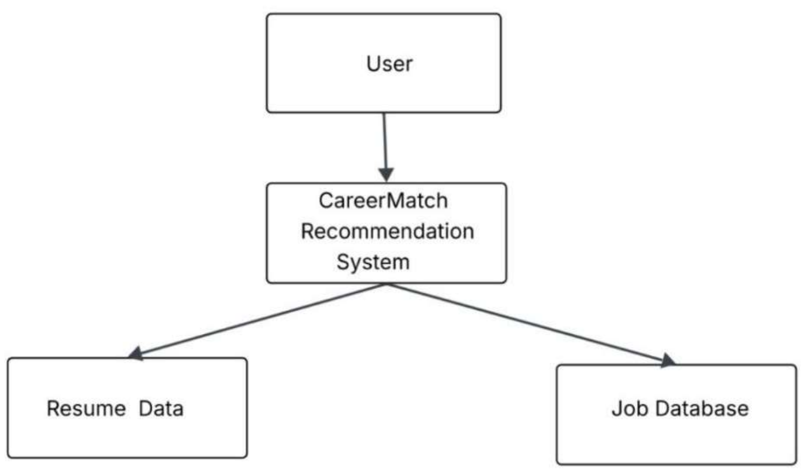

Figure 3.3 – DFD Level 0

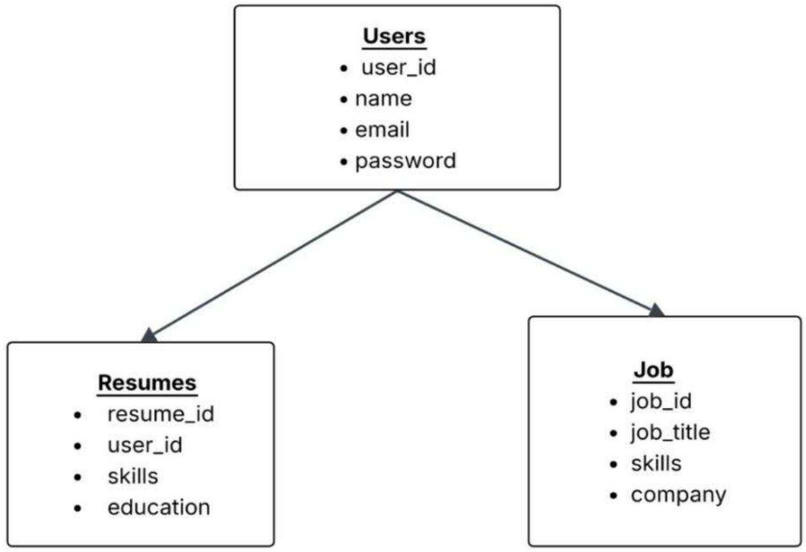

Figure 3.4 – Entity Relationship Diagram (ER Diagram)

CHAPTER 4

RESULTS AND DISCUSSION

4.1 System Implementation

CareerMatch is a program that has been used to assist the new graduates in getting appropriate employment opportunities depending on their qualifications. The system is based on frontend technologies such as HTML, CSS and JS to develop dynamic interface and PHP as server-side programming language. The data is saved in MySQL database where data can be managed effectively.

The implementation of the system is focused on automating the job search process by automatic resume processing and job matching. Registered users have an opportunity to post their resumes and receive job recommendations, depending on the skills they have identified in their resumes. The system analyses the resumes and matches them with job descriptions in the database.

The application of the CareerMatch system demonstrates how web technologies and machine learning algorithms can be combined to create a successful job matching system.

4.2 Functional Results

Jobs recommendation system was tried using different user profiles and resumes. The system was put to test by uploading resumes which contained varying technical skills and viewing the job recommendations.

The system was able to introduce the following functions successfully:

1. User-Registration and Login

The system enables users to sign-up and securely authenticate themselves.

2. Upload and Processing Resumes

The users may put in their resumes in a diverse variety of formats. It then scans the resume and picks up keywords and skills.

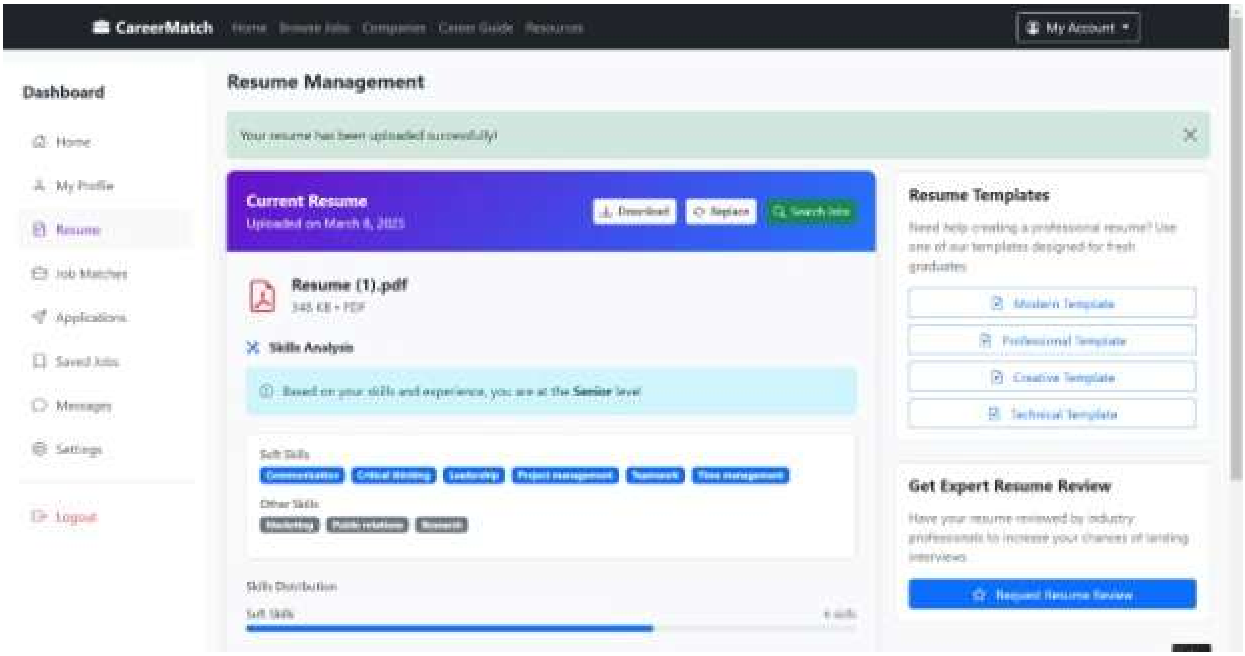

Figure 4.1-Resume Upload and Skills Extraction

3. Skill Extraction

The essential technical skills are removed and stored to a database.

Table 4.1: Sample Resume Skill Extraction

Resume ID	Extracted Skills	Education
R001	Python, Machine Learning, SQL	B.Tech Computer Science
R002	Java, Spring Boot, MySQL	B.Tech Information Technology
R003	HTML, CSS, JavaScript, React	B.Tech Computer Science

1. **Job Recommendation Generation**

The recommendation engine will compare the skills extracted with job descriptions and create a list of job opportunities that are relevant.

2. **Job Listing Management**

The administrator is able to add and manipulate job postings in the system database.

These features affirm that the system is able to carry out automated job matching based on the profile of the candidate.

4.3 Sample Output

The Bot reads through the content of the resume to find skills and matches them with job descriptions. We are going to assume that a candidate has Python, Machine Learning, and Data Analysis as the resume skills.

It then matches the profile against role descriptions to offer a variety of job roles. These roles may entail:

- Data Analyst
- Machine Learning Engineer
- Software Developer
- Backend Developer

The jobs that are recommended will appear on the user's dashboard so they can apply for a job they may be interested in.

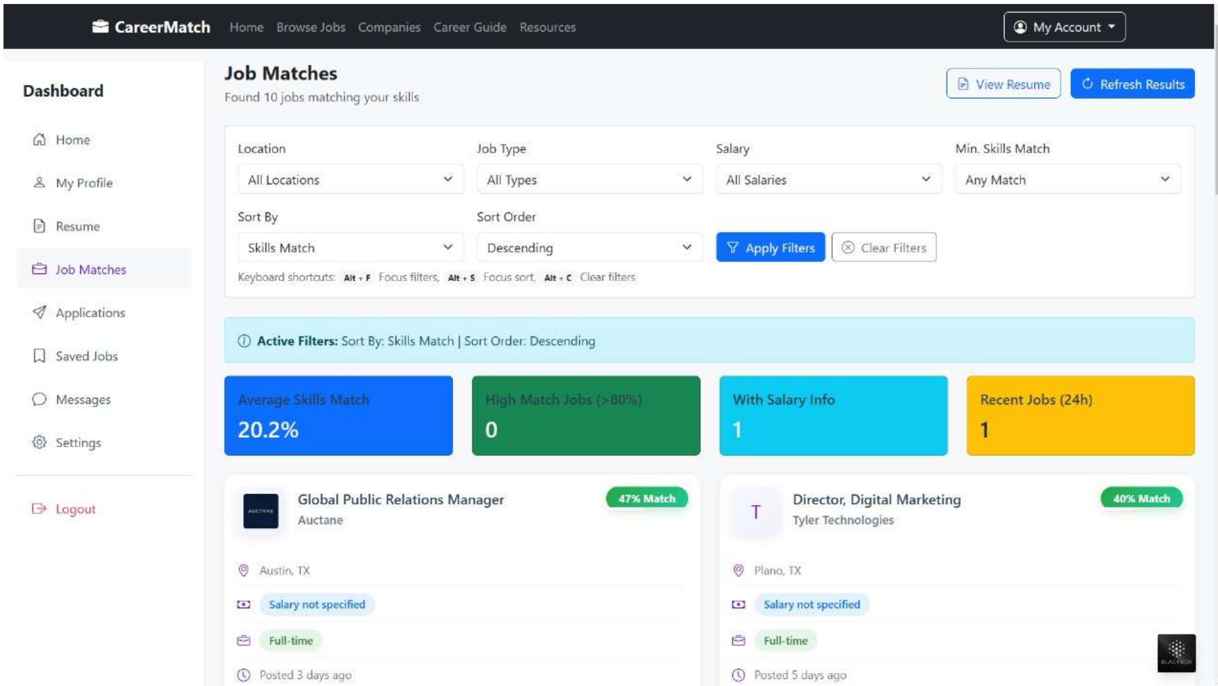

Figure 4.2- Job Matches Results

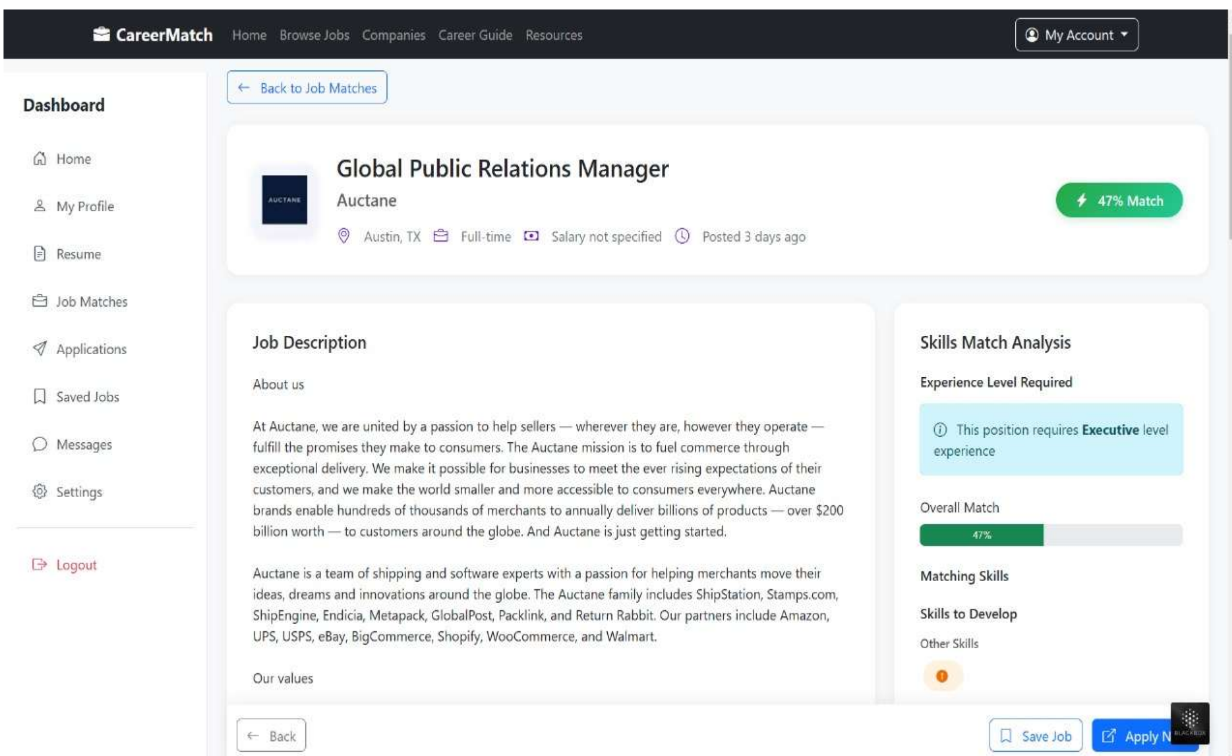

Figure 4.3-Job Description and Skills Match Analysis

Table 4.2: Job Recommendation Results

Candidate ID	Key Skills	Recommended Job Role
C101	Python, Data Analysis	Data Analyst
C102	Java, Backend Development	Backend Developer
C103	HTML, CSS, JavaScript	Frontend Developer

4.4 Discussion

Through implementation, the results obtained indicate that the CareerMatch platform is effective in helping users in finding job opportunities, which match their skill sets. The system saves time used in conducting manual job searches because it automates the process of analyzing resumes and matching them with available jobs.

Accuracy of the process of matching resumes to job descriptions is enhanced by the use of text analysis techniques such as TF-IDF. This method will help system to see opportunities of relevant jobs even when the words used in the resumes and job descriptions are slightly different.

The other benefit of the system is that the system has a user-friendly interface that enables users to easily post the resumes and see which job roles they are recommended. The platform also allows easier updating of job listings and is more easily maintained due to its modular design.

Nevertheless, there are some shortcomings in the present application. Recommendations are accurate depending on the quality of job data stored in the system and the completeness of the information provided in the resume of user. Increasing number of datasets and incorporating job API in real time could further enhance performance of the system.

In general, findings reveal that CareerMatch system is effective in proving that an intelligent job recommendation system can be successfully developed and implemented to fresh graduates.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The CareerMatch project, an online job recommendation system aimed at helping fresh graduates to select the right job opportunities based on their skills, education level and their career interests. The present-day job market is fraught with problems as many graduates struggle to find the right job options in the current job market since there are a lot of job opportunities in the job market and there is a lack of personalized guidance. One way of solving this problem is by using the CareerMatch system that will provide an intelligent platform that will analyse user resumes and will recommend job roles that will best fit the profile of the user.

The system involves integration of the web-development technologies and text analysis techniques in order to automate the job matching process. Allowing users to post their resumes, the platform will be able to extract their important skills and compare them with job descriptions stored in the database. Applying text processing methods like TF-IDF can help the system to recognize the key words and calculate the resemblance between the job specifications and the candidate profiles.

The adoption of the system proves that an automated job recommendation system can help to make the job search process much easier. The application developed has a user-friendly interface, resume processing, and personalized job recommendations. The system is also designed in a modular way that enables easy managing of job listing and user data.

Generally, CareerMatch platform can be used to bridge the gap between fresh graduates and potential employers by suggesting relevant job opportunities based on the skills of fresh graduates. The project points out the potential of using machine learning and web technologies together to develop intelligent systems of recruitment support.

5.2 Future Scope

Even though the present iteration of the CareerMatch platform fully achieves the resume analysis and job recommendation, the platform can still be expanded and improved.

The addition of real-time job APIs of large job portals could be one of the possible improvements. This would enable the system to automatically fetch the updated job listing on external platforms and offer more opportunities to the users.

The other solution might be adopting improved ML or deep learning algorithms to enhance the accuracy of job suggestions. More advanced algorithms might be used to examine the profiles of the candidates in more detail and to determine the complex dependencies between the skills and the job demands.

It was also possible to extend the system to cover career guidance functionalities, including skill gap analysis and course recommendations. This would assist the users to determine the skills they should acquire to qualify to particular job roles.

Moreover, creating a mobile application version of CareerMatch would make the platform more available to those users that prefer to use smartphones to search jobs. It may also be mobile-integrated to ensure that its features include push notifications of new job suggestions.

Additional features of the system, including interview preparation tools, company insights and networking options could also be added to the system in the future to further empower job seekers in their career development.

By making such improvements, the CareerMatch platform can potentially become a full-fledged career support system that will not only suggest jobs, but also assist users in developing the skills necessary to successfully be employed.

REFERENCES

[1] *A general graph learning based framework for job offer recommender system*. (2025). *Engineering Applications of Artificial Intelligence*.

[2] Arun Kumar, J. R., et al. (2025). Job and internship recommendation system. *Journal of Scientific Computing and Engineering Research (JSCER)*, 8(5).

[3] Babu, C. R. K., Kumar, T. S. R., & Sumith, S. (2025). Personalized job search with AI: A recommendation system integrating real-time data and skill-based matching. *International Journal of Engineering Research & Technology (IJERT)*, 14(4).

[4] Charan, C. S. N., Suhas, G. K., Yathisha, Y. L., & D. S. N. (2025). Job recommendation system using content-based and collaborative filtering. *Journal of Information Systems Engineering and Management (JISEM)*.

[5] Foo, B. L., Ting, C. Y., Goh, H. N., Quek, A., & Cham, C. L. (2025). Dynamic job recommendation by profiling undergraduates' academic performances. *Journal of Information and Web Engineering (JIWE)*.

[6] Jiang, Y. (2025). Design of dynamic career path recommendation system based on Markov decision process. *Journal of Computational Methods*. Sage Publications.

[7] Kanwal, T., & Amjad, T. (2024). Research paper recommendation system based on multiple features from citation network. *Scientometrics*. Springer.

[8] Kwieciński, R., Filipowska, A., Górecki, T., & Dubrov, V. (2023). Job recommendations: Benchmarking of collaborative filtering methods for classifieds. *arXiv*.

[9] Ndolo, D. M. (2023). Job recommendation systems: A literature review. *International Journal of Innovative Science and Research Technology (IJISRT)*, 8(2).

[10] Ong, X. Q., & Lim, K. H. (2023). SkillRec: A data-driven approach to job skill recommendation for career insights. *arXiv*.

[11] Parkar, A., Deshmukh, M., & Deshmukh, G. (2025). Job recommendation system. *International Journal of Basic and Applied Sciences (IJBAS)*.

[12] Soni, S., & Shah, S. (2025). Job recommendation systems through AI and machine learning. *International Journal of Basic and Applied Sciences*.

[13] Wang, S. (2025). Social network analysis enhances job recommender system using PageRank algorithm. *Journal of Computational Methods*. Sage Publications.

[14] Yap, R. E., Haw, S. C., & Al-Juboori, S. (2025). A comprehensive review on machine learning-based job recommendation systems. *International Journal of Recent Advances in Sciences (IJORAS)*, 7(2).

[15] Zheng, Z., Qiu, Z., Hu, X., Wu, L., Zhu, H., & Xiong, H. (2023). Generative job recommendations with large language model. *arXiv*.

APPENDIX 1

A.1 Database Schema

The CareerMatch system relies on MySQL database to store user data, resumes & extracted skills and jobs. The following tables are used in the system.

Table A.1 Users Table

Field Name	DataType	Description
user_id	INT (Primary Key)	Unique identity of the user
name	VARCHAR(50)	Name of the user
email	VARCHAR(50)	Email address
password	VARCHAR(250)	Encrypted password
created_at	DATETIME	Account creation time

Table A.2 Resumes Table

Field Name	DataType	Description
resume_id	INT (Primary Key)	Unique identity of resume
user_id	INT (Foreign Key)	ID of the user
file_name	VARCHAR(250)	Uploaded resume file
skills	TEXT	Extracted skills from resume

Table A.3 Jobs Table

Field Name	DataType	Description
job_id	INT (Primary Key)	Unique job identity
job_title	VARCHAR(50)	Job role
company_name	VARCHAR(50)	Name of company
required_skills	TEXT	Skills required for job
location	VARCHAR(100)	Job location

Below is a sample PHP code used to upload resumes to the system.

```
<?php if(isset($_POST['upload'])){
        $file = $_FILES['resume']['name'];
        $temp = $_FILES['resume']['tmp_name'];

        $folder = "uploads/".$file; if(move_uploaded_file($temp,

        $folder)){
                echo "Resume uploaded successfully!";
        } else {
                echo "Failed!";
        }
}
?>
```

A.2 Sample Job Recommendation Logic

The system compares extracted resume skills with job descriptions using text similarity techniques.

Example:

- Resume Skills: Python, Machine Learning, SQL
- Job Requirement: Python, Data Analysis, SQL

Based on similarity between these keywords, the system recommends Data Analyst or Machine Learning related roles.

A.3 Project Repository

The complete source code of the CareerMatch platform is maintained in the project repository which contains:

- Project Report
- Source Code
- Presentation Slides
- Machine Learning

Scripts Repository

Structure:

```
CareerMatch
│
├── frontend
├── backend
├── database
├── machine_learning
└── documentation
```

RESEARCH PAPER/PATENT

In the final year project, entitled CareerMatch: A Smart Job Recommendation System, a patent application is made successful submission of application to the Office of the Controller General of Government of India under the Department of Promotion of: Patents, Designs and Trade Marks.

Industry and Internal Trade (DPIIT), Ministry of Commerce & Industry.

The proposed system is a smart platform which uses the resumes and matches of various users to offer a match.

them with appropriate job opportunities depending on the extracted skills, qualifications and preferences.

The system combines resume parsing, skill mining, and machine learning-powered.

recommendation methods to enhance the efficiency of job search process of fresh.

graduates.

The invention aims at filling gap between job-seekers and companies by offering.

individualized suggestions via computerized resume examination.

Patent Application Details

Title of Invention: CareerMatch: A Smart Job Recommendation System

Application Number: 202511108555

Application Type: Ordinary Application

Date of Filing: 08 November 2025

Applicants:

1. Dr. Payal Chhabra
2. Mr. Anand Sharma
3. Mr. Harsh Vikram Singh
4. Mr. Prabal Gupta
5. Ms. Jyoti Kumari

The report has the attached proof of submission of patent application.

PROOF OF SUBMISSION/ACCEPTANCE/PUBLICATION

The figure below indicates the recognition or evidence of filing of the patent application on the CareerMatch system, which proposes an intelligent job recommendation system to fresh graduates through the analysis of their resumes and machine learning algorithms.

The patent submission demonstrates the novelty of proposed system and its potential contribution to the field of smart recruitment and career guidance systems.

सत्यमेव जयते

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Department for Promotion of Industry and Internal Trade
Ministry of Commerce & Industry,
Government of India

INTELLECTUAL PROPERTY INDIA
PATENTS | DESIGNS | TRADE MARKS
GEOGRAPHICAL INDICATIONS

Application Details	
APPLICATION NUMBER	202511108555
APPLICATION TYPE	ORDINARY APPLICATION
DATE OF FILING	08/11/2025
APPLICANT NAME	**1 . Ms. Payal Chabra** **2 . Mr. Anand Sharma** **3 . Mr. Harsh Vikarm Singh** **4 . Mr. Prabal Gupta** **5 . Ms. Jyoti Kumari**
TITLE OF INVENTION	CareerMatch: A Smart Job Recommendation System
FIELD OF INVENTION	MECHANICAL ENGINEERING
E-MAIL (As Per Record)	patentpointservices@gmail.com
ADDITIONAL-EMAIL (As Per Record)	
E-MAIL (UPDATED Online)	
PRIORITY DATE	
REQUEST FOR EXAMINATION DATE	--
PUBLICATION DATE (U/S 11A)	26/12/2025

Figure A.1 Proof of Patent Application Submission

PLAGIARISM REPORT

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